the proposed system under realistic deployment scenarios.

Results

The Android application was successfully developed using a microservices architecture. It includes the following key components:

- A scraping service using FastAPI and Selenium for extracting product data from Zepto, Swiggy Instamart, BigBasket, and JioMart.
- A backend developed in Java for managing data, exposing REST APIs, and enabling comparison logic.
- The user must first enter the name of any product and search for it. The application will then show products from all the configured websites and sort them from the lowest to the highest price as shown in Figure 3.
- If a user clicks on any product, the user will be redirected to the actual app (if available) or website to see more details as shown in Figure 7.
- There are two floating buttons. The first one gives an option to filter out products on the basis of the source (Zepto, Swiggy, etc) as shown in Figure 8.
- The second floating button takes you to the 'Comparison list' screen, which gives the total value of products from each source. Users can choose the source as per the total value and shop from a particular website (Figure 9).

The modular structure of this system makes it easy to integrate new retailers over time, without increasing maintenance complexity. Notably, the implementation follows ethical scraping guidelines, ensuring minimal server load and respectful adherence to each platform's usage policies.

Beyond its technical merits, the system provides real-world value by helping users make quicker, better informed shopping decisions. In a highly price-sensitive market, real-time comparisons empower users to save both money and the effort of switching between multiple apps. This is how open source technologies, when

Figure 7: Redirection to source platform for selected product details after price comparison